Weekly Report - Week 7

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Branch: Feat-3.2-AssessmentContex

Completed Tasks

https://github.com/SWEN90017-2025-SM1/OAS-web-based-oral-assessments/commit/da4ebec391a189ae766c88643c9c36aa2515abdc

https://github.com/SWEN90017-2025-SM1/OAS-web-based-oral-assessments/commit/cc769fcb506cd632280d0b6f5d26bec8c8ecd333

https://github.com/SWEN90017-2025-SM1/OAS-web-based-oral-assessments/commit/859d76ea25a35922e45653340b439ebcdf7ab6f7

Major Schema Migration and Service Refactoring

This week focused on implementing critical backend refactoring to address deprecated database schema components and ensure system compatibility with the new architectural design.

1. MarkingServiceImpl Complete Refactoring

- **Deprecated Dependencies Removal**: Eliminated all references to `time_slots` and `time_slot_examiners` tables, which are being phased out in favor of the new session-based architecture
- **Session-Examiner Integration**: Migrated examiner assignment logic from `TimeSlotExaminersMapper` to `SessionExaminerMapper`, ensuring proper examiner validation through the `session_examiners` table
- **Venue Information Restructuring**: Updated venue retrieval logic to source venue details directly from `SlotSession` entities rather than deprecated `TimeSlot` references
- **Assessment History Integration**: Implemented `buildExistingAssessment()` method to populate existing assessment data from `AssessmentSubmission` records, enabling proper assessment continuity in the marking interface

2. Entity-Subject Semantic Clarification

- **DTO Architecture Overhaul**: Refactored `AdminAssessmentSearchResponseDTO` by replacing `SubjectInfoDTO` with semantically accurate `EntityInfoDTO`, clarifying the distinction between course subjects and assessment entities (students/groups)
- **Field Renaming and Logic Correction**: Updated `groupMembers` field to `students` and corrected entity identification logic to properly handle both individual and group assessment scenarios
- **Database Query Optimization**: Modified SQL queries in `AssessmentSubmissionMapper.xml` to remove deprecated table joins and implement proper entity-session relationships

3. Hardcoded Test Data Implementation

- **Temporary Entity Mapping**: Implemented hardcoded entity IDs (student ID: 2L, group ID: 1L) with comprehensive TODO documentation for future dynamic mapping implementation
- **Booking System Compatibility**: Addressed the temporary gap between deprecated `bookedSubjectId` / `bookedSubjectType` fields and the upcoming multi-booking system by providing robust fallback mechanisms

Technical Challenges and Solutions

1. Complex Database Schema Transition

Challenge: Managing the transition from time-slot-based architecture to session-based architecture while maintaining system functionality required careful dependency mapping and potential circular reference resolution.

Solution: Implemented a systematic approach by first identifying all deprecated dependencies, then progressively replacing them with session-based alternatives. Used comprehensive logging and error handling to ensure smooth migration without data loss.

2. Entity-Session Mapping Gap

Challenge: The current system lacks a direct mapping mechanism between assessment sessions and their associated entities (students/groups), as the previous booking system is deprecated and the new system is still under development.

Solution: Designed a robust hardcoded fallback system with clear TODO documentation and architectural notes. This approach ensures immediate system functionality while providing a clear migration path for future implementation of dynamic entity-session mapping.

3. Assessment Continuity Implementation

Challenge: Integrating existing assessment data retrieval while working with hardcoded entity mappings required careful consideration of data consistency and proper JSON parsing for complex assessment structures.

Solution: Developed a comprehensive `buildExistingAssessment()` method that handles various assessment data formats and gracefully manages partial data scenarios. Implemented proper error handling for JSON parsing operations and provided meaningful fallbacks for missing assessment components.

Next Week Planning

1. Dynamic Entity-Session Mapping Implementation

- Collaborate with the booking system development team to implement proper dynamic mapping between sessions and entities
- Replace hardcoded entity IDs with database-driven entity resolution
- Implement comprehensive testing for various entity-session combination scenarios

2. Performance Optimization and Testing

- Conduct thorough performance testing of the refactored `MarkingServiceImpl` with real-world data volumes
- Optimize database queries for the new session-examiner relationship structure
- Implement integration tests for the complete marking workflow

3. Documentation and Code Review Preparation

- Complete comprehensive documentation for the new session-based architecture
- Prepare detailed migration notes for other team members working on related components
- Conduct code review sessions to ensure architectural consistency across the marking system

Summary

This week's work represents a significant milestone in the backend architectural migration, successfully removing deprecated dependencies while maintaining system functionality. The implementation of hardcoded fallbacks with clear migration paths ensures both immediate operational capability and future extensibility. The refactored codebase now aligns with the modern session-based architecture and provides a solid foundation for upcoming feature enhancements.